

Introduction

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1.1 Designed for Safety Applications

The RM46x device architecture has been designed from the ground up to simplify development of functionally safe systems. The basic architectural concept is known as a safe island approach. Power, clock, reset, and basic processing function are protected to a high level of diagnostic coverage in hardware. Some of the key features of the safe island region are:

- A dual core lockstep processing solution built around ARM® Cortex®-R4F CPU that detects failures at the core boundary on a cycle by cycle basis. Special measures in processor layout, clock distribution, power distribution, reset distribution, and temporal diversity are all implemented to mitigate common cause failures of the logical CPU and its checker. For complete details on the ARM® Cortex®-R4F CPU, refer to the [ARM® Cortex®-R4F Technical Reference Manual](#).
- Hardware BIST controllers which provide an extremely high level of diagnostic coverage for the lockstep CPUs and SRAMs in the system, while executing faster and consuming less memory than equivalent software-based self-test solutions
- ECC on the SRAM and flash memories tightly coupled to the R4F. The ECC controllers are located inside the CPU. This approach has two key advantages:
 - The interconnect between CPU and the memory is also covered by the diagnostic
 - The ECC logic itself is checked on a cycle by cycle basis
- Onboard voltage and reset monitoring logic
- Onboard oscillator and PLL failure detection logic including a backup RC oscillator which can be utilized upon failure

The RM46x device architecture also includes many features to simplify diagnostics of remaining logic such as:

- Continuous parity diagnostics on all peripheral memories
- Analog and digital loopback to test for shorts on I/O
- HW self-test and diagnostics on the ADC module to check integrity of both analog inputs and the ADC core conversion function
- A DMA driven hardware engine for the background calculation of CRC signatures during data transfers
- A centralized error reporting function including a status output pin to enable external monitoring of the device status

1.2 Family Description

The RM46x integrates the ARM® Cortex®-R4F Floating Point CPU that offers an efficient 1.6 DMIPS/MHz and has configurations that can run up to 220MHz providing up to 352 DMIPS. The device supports the little-endian [LE] format.

The RM46x has up to 1.25MB integrated Flash and up to 192KB data RAM configurations with single bit error correction and double bit error detection. The flash memory on this device is a nonvolatile, electrically erasable and programmable memory implemented with a 64-bit-wide data bus interface. The flash operates on a 3.3V supply input (same level as I/O supply) for all read, program and erase operations. When in pipeline mode, the flash operates with a system clock frequency of up to 220MHz. The SRAM supports single-cycle read/write accesses in byte, halfword, and word modes.

The RM46x device features peripherals for real-time control-based applications, including two Next Generation High End Timer (N2HET) timing coprocessors with up to 40 total IO terminals and a 12-bit A to D converter supporting up to 24 inputs.

The N2HET is an advanced intelligent timer that provides sophisticated timing functions for real-time applications. The timer is software-controlled, using a reduced instruction set, with a specialized timer micromachine and an attached I/O port. The N2HET can be used for pulse width modulated outputs, capture or compare inputs, or general-purpose I/O. It is especially well suited for applications requiring multiple sensor information and drive actuators with complex and accurate time pulses. A High End Timer Transfer Unit (HET-TU) can perform DMA type transactions to transfer N2HET data to or from main memory. A Memory Protection Unit (MPU) is built into the HET-TU.

The device has two 12-bit-resolution MibADCs with 24 total channels and 64 words of parity protected buffer RAM each. The MibADC channels can be converted individually or can be grouped by software for sequential conversion sequences. Sixteen channels are shared between the two MibADCs. There are three separate groupings. Each sequence can be converted once when triggered or configured for continuous conversion mode.

The device has multiple communication interfaces: three MibSPIs, up to two SPIs, one LIN, one SCI, three DCANs, one I²C, one Ethernet, and USB Host and Device modules. The SPI provides a convenient method of serial interaction for high-speed communications between similar shift-register type devices. The LIN supports the Local Interconnect standard 2.0 and can be used as a UART in full-duplex mode using the standard Non-Return-to-Zero (NRZ) format. The DCAN supports the CAN 2.0B protocol standard and uses a serial, multimaster communication protocol that efficiently supports distributed real-time control with robust communication rates of up to 1 megabit per second (Mbps). The DCAN is ideal for applications operating in noisy and harsh environments (for example, automotive and industrial fields) that require reliable serial communication or multiplexed wiring. The Ethernet module supports MII and MDIO interfaces.

The I2C module is a multi-master communication module providing an interface between the microcontroller and an I2C compatible device via the I2C serial bus. The I2C supports both 100 Kbps and 400 Kbps speeds.

The frequency-modulated phase-locked loop (FMPLL) clock module is used to multiply the external frequency reference to a higher frequency for internal use. The FMPLL provides one of the seven possible clock source inputs to the global clock module (GCM). The GCM module manages the mapping between the available clock sources and the device clock domains.

The device also has an external clock prescaler (ECP) module that when enabled, outputs a continuous external clock on the ECLK pin/ball. The ECLK frequency is a user-programmable ratio of the peripheral interface clock (VCLK) frequency. This low frequency output can be monitored externally as an indicator of the device operating frequency.

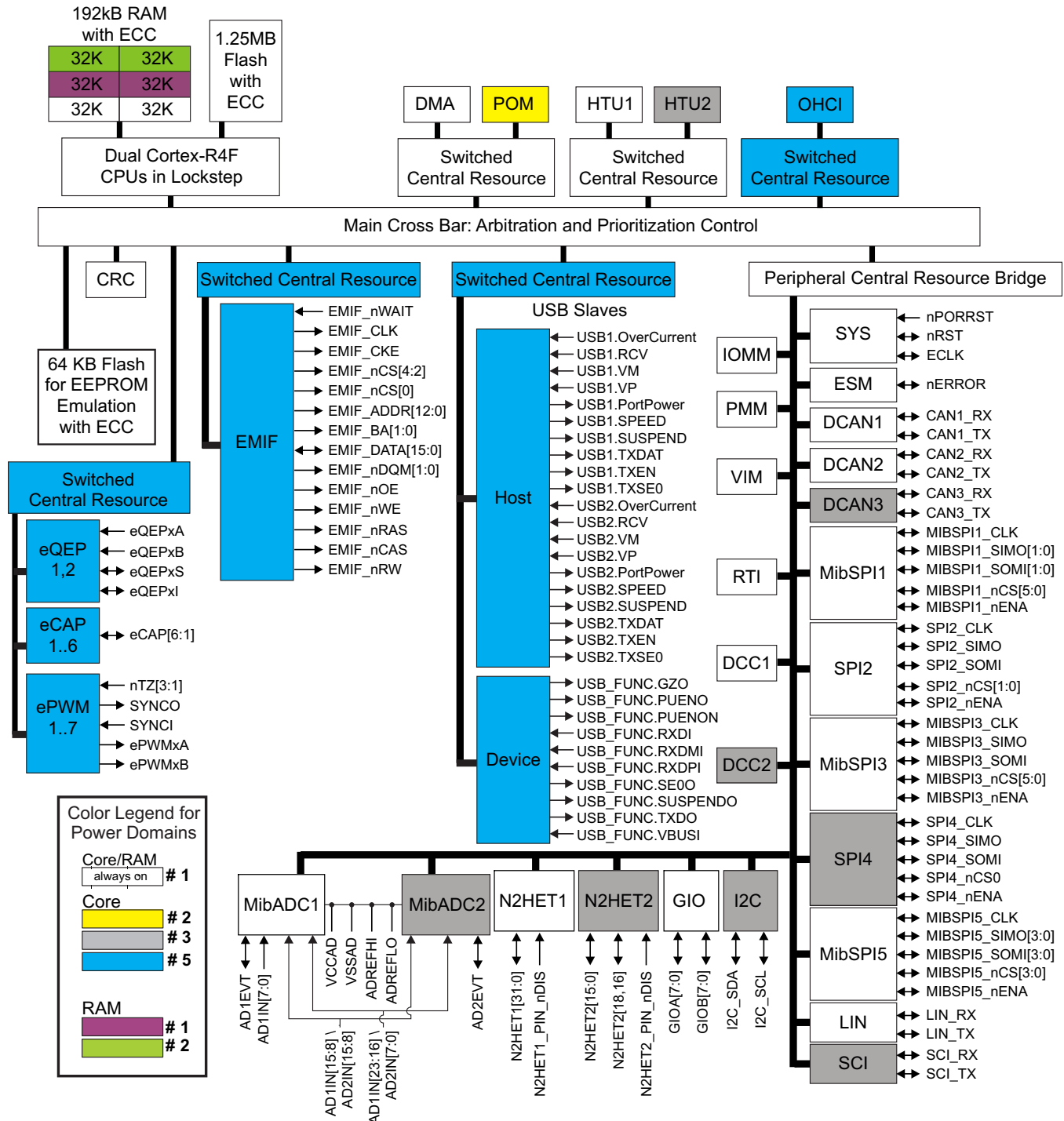
The Direct Memory Access Controller (DMA) has 16 channels, 32 control packets and parity protection on its memory. A Memory Protection Unit (MPU) is built into the DMA to protect memory against erroneous transfers.

The Error Signaling Module (ESM) monitors all device errors and determines whether an interrupt or external Error pin/ball is triggered when a fault is detected. The nERROR can be monitored externally as an indicator of a fault condition in the microcontroller.

The External Memory Interface (EMIF) provides a memory extension to asynchronous and synchronous memories or other slave devices.

The device supports all the built-in ARM® Cortex®-R4F CoreSight™ debug features. In addition, a Parameter Overlay Module (POM) can re-route Flash accesses to internal memory or to external memory via the EMIF, thus avoiding the re-programming steps necessary for parameter updates in Flash.

With integrated safety features and a wide choice of communication and control peripherals, the RM46x is an ideal solution for high performance real time control applications with safety critical requirements.

Figure 1-1. Block Diagram


1.3 Endianism Considerations

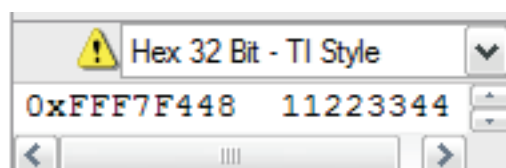
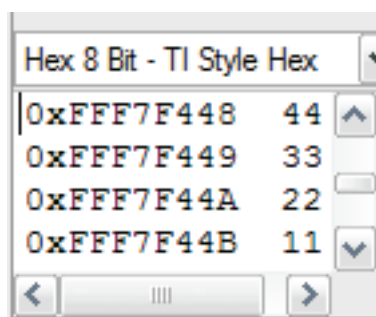
1.3.1 RM46x: Little Endian (LE)

The RM46x family is based on the ARM® Cortex®-R4F core. ARM has designed this core to be used in big-endian and little-endian systems. For the TI RM46x family, the endianness has been configured to little-endian. Little-endian systems store the least-significant byte of a multi-byte data field in the lowest memory address. Also, the address of the multi-byte data field is the lowest address. Following is an example of the physical addresses of individual bytes.

Figure 1-2. Example: SPIDELAY – 0xFFFF7F448

31	24	23	16
C2TDELAY[7:0]		T2CDELAY[7:0]	
Byte 3 - 0xFFFF7F44B		Byte 2 - 0xFFFF7F44A	
15	8	7	0
T2EDELAY[7:0]		C2EDELAY[7:0]	
Byte 1 - 0xFFFF7F449		Byte 0 - 0xFFFF7F448	

32-bit accesses to this register should use the lowest address, that is, 0xFFFF7F448. Writing 0x11223344 to address 0xFFFF7F448 shows the following when viewing the memory in 8-bit and 32-bit modes.



As such, the headers provided as part of HALCoGen do take the endianness into account and provide header structures that are agnostic to endianness. This is achieved by using C directives for the compiler that make use of the compile options configured for the project by the user (`__little_endian__` used in Code Composer Studio codegen tools). This directive may need to be adapted for other compilers.

```
#ifdef __little_endian__
    char C2DELAY      : 8U; /**lt; 0xF448: CS to ENA      */
    char T2DELAY      : 8U; /**lt; 0xF449: Transmit to ENA */
    char T2CDELAY      : 8U; /**lt; 0xF44A: Transmit to CS  */
    char C2TDELAY      : 8U; /**lt; 0xF44B: CS to Transmit */
#else
    char C2DELAY      : 8U; /**lt; 0xF448: CS to Transmit */
    char T2CDELAY      : 8U; /**lt; 0xF449: Transmit to CS  */
    char T2EDELAY      : 8U; /**lt; 0xF44A: Transmit to ENA */
    char C2EDELAY      : 8U; /**lt; 0xF44B: CS to ENA      */
#endif
```